Bitana V6 â€” Complete System Audit & Evolution Report **Document Version:** 1.0 **Date:** 2026-05-25 **Author:** OWL (for Martin) **Repository:** `github.com:vsehuinya1/bitana` **Branch:** `main` **Latest Tag:** `v6.4` --- ## Table of Contents 1. [Executive Summary](#1-executive-summary) 2. [System Architecture Overview](#2-system-architecture-overview) 3. [Full Version History & Git Log](#3-full-version-history--git-log) 4. [What Worked â€” What Didn't](#4-what-worked--what-didnt) 5. [Known Bugs & Issues](#5-known-bugs--issues) 6. [Live Performance Analysis](#6-live-performance-analysis) 7. [Confirmation Gate Analysis](#7-confirmation-gate-analysis) 8. [Stop Loss Analysis](#8-stop-loss-analysis) 9. [Recommendations for Next AI](#9-recommendations-for-next-ai) 10. [Current Configuration Reference](#10-current-configuration-reference) --- ## 1. Executive Summary Bitana is a liquidation-cascade breakout trading system for Binance USDT-M perpetual futures, executing on 5m candles. It evolved from V3 through V6.4 over approximately 2 weeks of intensive development. **Current state (v6.4):** - 149 closed trades, -28.25R total, 45% WR - 57 symbols (35 proven + 22 experimental) - 25/57 cascades currently active - System is running but has **negative expectancy** (-0.19R/trade) - A **critical bug** exists: the `imb` confirmation gate passes 83% of the time despite `imb_z=0.00` on every entry â€” the gate is effectively a free pass - The `breakout` gate fires 0% of the time â€” no trade in the last 18 entries broke above the 60-bar range high **The system was directionally correct at one point.** V6.2 (before the confirmation loosening) showed promising results on D5 and D8, and the early-cut mechanism was saving value. The loosening in V6.4 was intended to increase entry frequency but introduced low-quality entries that are bleeding on stop losses. **Key finding:** The entry problem is not the thresholds â€” it's that the `imb` gate is broken and `breakout` never fires. The system is effectively trading on `body + momentum` alone (3/6 gates that almost always pass). --- ## 2. System Architecture Overview ### Strategy: 3-Play Liquidation Cascade Breakout | Play | Description | |------|-------------| | A | Breakout continuation after liquidation cascade | | B | Liquidation cascade capture (momentum ignition) | | C | Momentum ignition after reclaim | ### Core Components | File | Purpose | |------|---------| | `engines/liq_cluster_engine_v5.py` | Main trading engine â€” entry/exit logic, confirmations, cascade detection | | `tools/v5_forward_test.py` | Live runner â€” WebSocket, candle loop, position management, Telegram alerts | | `config/v5_forward_test.yaml` | Symbol configuration (tiers, risk, cascade params) | | `data/candle_manager.py` | Candle creation, taker_buy_volume population | | `v6_telemetry.py` | Async telemetry â€” R-path tracking, shadow exits, exit attribution | | `shadow_exits.py` | Shadow exit evaluator (research) | ### Data Sources | Source | Role | Status | |--------|------|--------| | Binance WebSocket `!forceOrder@arr` | Primary liquidation feed | âœ… Active | | Coinalyze DB | Historical liq seeding (cold start) | âœ… 90 symbols | | Binance REST (klines) | 5m candle data | âœ… Active | | Binance REST (aggTrade) | taker_buy_volume | âš ï¸� Often zero | ### Infrastructure - **VPS:** 161.97.185.65 (root, systemd managed) - **Service:** `bitana-v5-paper.service` - **DB:** SQLite WAL mode (`v5_forward_test.db` + `v6_telemetry.db`) - **Telemetry:** Async queue + background thread pool + fallback spooling - **Alerts:** Telegram bot (HTML mode, 2-attempt retry, 4000 char truncation) --- ## 3. Full Version History & Git Log ### v5.0 â†’

v5.3 (Pre-audit baseline) | Tag | Date | Commit | Description | |-----|------|--------|-------------| | v5.2 | 2026-05-22 | `ab5237c` | Migrate Coinalyze â†’ Binance allForceOrders + disable squeeze filter | | v5.3 | 2026-05-23 | `ffb2f76` | Migrate REST endpoint to WebSocket `!forceOrder@arr` for zero rate-limiting | **What happened:** Coinalyze API deprecated `allForceOrders` (400 error). Migrated to Binance WebSocket as primary liq feed. Coinalyze kept for cold-start seeding only. ### V6.0 â€" Full Audit Fix (2026-05-23) | Tag | Date | Commit | Description | |-----|------|--------|-------------| | v6.0 | 2026-05-23 | `cb03ba7` | Full audit fix â€" 18 bugs fixed | **Engine fixes (9):** 1. **P0** Removed double `bars_held` increment in `manage_position()` â€" winners were exiting at 2x speed (D1/D2 trades with max_hold=500 exiting after ~250 bars) 2. **P1** Fixed `imb_z` to use real taker buy imbalance z-score (was a meaningless price-distance metric) 3. **P1** Widened aggression mapping from [-2,+2] to [-3,+3] â€" **THIS WAS A REGRESSION** (see below) 4. **P2** `stop_cooldown = 288` (24h) instead of 999999 (permanent ban) 5. **P2** Time-based stop_cooldown decrement in `evaluate()` 6. **P2** Cascade-deactivation exit tightening (1.0 ATR trail) 7. **P2** `get_risk_pct()` returns vol-targeted risk (was returning stale flat 4%) 8. **P3** Renamed `oi_acceleration` â†' `vol_acceleration` 9. **P3** Updated docstring to V6 **Runner fixes (9):** 1. **P0** Debounced WS engine updates (60s intervals, not per-event) 2. **P0** Batch SQLite commits + WAL mode 3. **P1** `asyncio.Lock` for engine state 4. **P2** Fixed equity snapshot position count 5. **P3** `deque(maxlen=200)` for candle buffers 6. **P3** Fixed duplicate key (removed hardcoded LONG) 7. **P3** Self-test checks all symbols 8. **P3** Fixed startup Telegram message 9. **P3** Updated all alert messages to V6 **Martin's assessment:** This was a necessary and mostly correct release. The aggression widening to [-3,+3] was the only mistake â€" it flooded D1 with marginal signals. ### V6.1 â€" Revert Aggression + Add Cascade Imb Gate (not tagged) | Date | Change | Description | |------|--------|-------------| | 2026-05-24 | Reverted aggression | [-3,+3] â†' [-2,+2] (Claude's widening was a regression) | | 2026-05-24 | Added `min_cascade_imb=0.30` | Filter neutral cascades â€" require directional imbalance | **Martin's assessment:** The aggression revert was correct. The `min_cascade_imb` gate was too aggressive â€" it blocked 11 out of 12 cascade-active symbols, causing 6+ hours of zero entries. ### V6.2 â€" D3 Rejection + Directional Filter + Early Cut (2026-05-24) | Tag | Date | Commit | Description | |-----|------|--------|-------------| | v6.2 | 2026-05-24 | `dde5950` | Reject D3, D1-D2 directional filter, early-cut dead trades | **Changes:** 1. **D3 rejection:** 6 trades, -3.91R, none with MFE>0.61R â€" added to rejection list with D4/D10 2. **D1-D2 directional filter:** Must pass `imb_z` OR `vol_z` confirmation 3. **Early cut:** At `decay_start_bar`, if `R < -0.3` AND `MFE < 0.3`, cut the trade **Martin's assessment:** This was the **best version**. D3 rejection was data-driven. Early cut was saving value. D1-D2 directional filter was reasonable. Results were directionally correct â€" D5 showed +1.81R on 4 trades (75% WR), D8 showed +2.91R on 12 trades (58% WR). The system was finding edges. ### V6.2 Telemetry Deployment (2026-05-24) | Date | Commit | Description | |------|--------|-------------| | 2026-05-24 | `2987ce9` | Deploy v6.2 research telemetry, shadow exits, infra fixes | | 2026-05-24 | `d91b670` | Async event-loop safe bounded queue-based telemetry isolation | | 2026-05-24 | `ae4cb8b` | Fix `is_experimental` in trades table + telemetry + close_position | ### V6.3 â€" Tier 2

Experimental Pairs (2026-05-24) | Tag | Date | Commit | Description | |-----|------|--------|-------------| | v6.3 | 2026-05-24 | `230c984` | Tier 2 experimental pairs (15 new, 50 total) | **Changes:** - Added 15 experimental pairs: HYPE, DOGE, SUI, GRASS, ONDO, LINK, AVAX, INJ, AAVE, PENGU, LTC, TRUMP, FARTCOIN, VIRTUAL, TIA - `is_experimental` column added to both `trade_entries` and `trades` tables - Runner patched to read `tier_c_experimental` ### V6.3 Fixes (2026-05-25) | Date | Commit | Description | |------|--------|-------------| | 2026-05-25 | `c4550ee` | Fix exit timestamp -5m offset bug (guard in `_manage_positions`) | | 2026-05-25 | `971bf2d` | Widen early_cut threshold to `R < -0.5` and `MFE < 0.1` | **Early cut analysis (Gemini trajectory analysis):** - 15 early_cut trades analyzed with actual 5m candle trajectories - Widening from `R<-0.3/MFE<0.3` to `R<-0.5/MFE<0.1` saves +2.2R to +3.1R - 8 trades survive the wider threshold (4 go on to win big: NEAR +4.33R, VIRTUAL +2.92R, ZEC +3.14R, AVAX +2.20R; 4 hit SL) - 7 still cut â€" net improvement +2.26R to +3.11R ### V6.3.1 â€" Liq Lookback Revert (2026-05-25) | Date | Commit | Description | |------|--------|-------------| | 2026-05-25 | `1842eef` | Shorten daily liq lookback to 30 days (MISTAKE) | | 2026-05-25 | `d8864fc` | **REVERT:** Restore liq_lookback to 90 days | **What happened:** Attempted to help dead symbols cascade by shortening lookback. Made things WORSE â€" lost 5 active cascades (28â†'23) because shorter window keeps recent spikes MORE prominent. Reverted within hours. ### V6.3.2 â€" Remove min_cascade_imb Gate (2026-05-25) | Date | Commit | Description | |------|--------|-------------| | 2026-05-25 | `b2bfabf` | Disable min_cascade_imb gate â€" p90 check is sufficient | **What happened:** The `min_cascade_imb=0.30` gate (added in V6.1) was blocking 11/12 cascade-active symbols. Removed it â€" p90 is now the sole cascade filter. This was the right call. ### V6.4 â€" Loosen Entry Confirmations (2026-05-25) â†� CURRENT | Tag | Date | Commit | Description | |-----|------|--------|-------------| | v6.4 | 2026-05-25 | `456dbed` | Loosen entry confirmations | **Changes:** | Parameter | Old | New | Rationale | |-----------|-----|-----|-----------| | `vol_z_threshold` | 3.0 | 2.0 | Above-avg volume, not extreme | | `imb_z_threshold` | 2.0 | 1.0 | Just needs positive taker pressure | | `body_strength_min` | 0.60 | 0.50 | Slightly more lenient | | `impulse_min_pct` | 0.30 | 0.20 | Lower min move | | `min_confirmations` | 4/6 | 3/6 | More entries, noisier | **Martin's assessment:** This was a mistake. The loosening increased entry frequency but the signal quality dropped significantly. 18 trades fired post-loosen, ALL would have been blocked by old rules. Result: -11.16R on 18 trades (22% WR both proven and experimental). The `imb` gate is broken (passes 83% with imb_z=0.00) and `breakout` never fires (0/18). --- ## 4. What Worked â€" What Didn't ### âœ… What Worked | Feature | Evidence | |---------|----------| | **D5 decile** | 4 trades, +1.81R, 75% WR â€" clean edge, needs more data | | **D8 decile** | 12 trades, +2.91R, 58% WR â€" solid performer | | **D3 rejection** | Zero D3 trades post-V6.2, was -4.38R on 9 trades before | | **Early cut (widened)** | Saving +2.2R to +3.1R vs old threshold â€" mechanically correct | | **Cascade deactivation exit** | 1 trade, +1.42R â€" tightened trail when cascade dies | | **WebSocket migration** | Zero rate-limiting, real-time liq aggregation working | | **WAL mode + batch commits** | Eliminated IO contention | | **WS debounce (60s)** | Eliminated per-event CascadeTracker rebuild | | **DASH pair** | 11 trades, +2.76R, 64% WR â€" best proven pair | | **SOL pair** | 12 trades, +0.77R, 58% WR â€" consistent | | **FET pair**

| 12 trades, +0.63R, 67% WR â€" steady | ### â�Œ What Didn't Work | Feature | Evidence | |---------|----------| | **Aggression widening [-3,+3]** | Flooded D1 with marginal signals â€" reverted | | **min_cascade_imb=0.30** | Blocked 11/12 cascade symbols â€" removed | | **liq_lookback=30** | Lost 5 active cascades â€" reverted to 90 | | **Confirmation loosening (V6.4)** | -11.16R on 18 trades, 22% WR â€" **REVERT RECOMMENDED** | | **D1 as a decile** | 72 trades, -15.67R, 47% WR â€" negative expectancy, but Martin says keep collecting data | | **D7 decile** | 13 trades, -3.34R, 31% WR â€" keep collecting per Martin | | **D9 decile** | 12 trades, -2.68R, 33% WR â€" keep collecting per Martin | | **LTC pair** | 4 trades, -2.84R, 0% WR â€" cut candidate | | **HYPE pair** | 3 trades, -2.14R, 0% WR â€" cut candidate | | **AVAX pair** | 4 trades, -2.25R, 25% WR â€" cut candidate | | **ZEC pair** | 12 trades, -5.29R, 33% WR â€" worst proven pair | | **NEAR pair** | 22 trades, -4.41R, 50% WR â€" high volume but negative | ### âš ï¸� Mixed / Needs More Data | Feature | Evidence | |---------|----------| | **D2 decile** | 18 trades, -4.55R, 44% WR â€" small sample, inconclusive | | **D6 decile** | 9 trades, -2.35R, 44% WR â€" small sample | | **Experimental pairs overall** | 30 trades, -9.06R, 30% WR â€" but VIRTUAL (+0.92R) and AAVE need individual evaluation | --- ## 5. Known Bugs & Issues ### ðŸ"´ CRITICAL: imb Confirmation Gate is Broken **File:** `engines/liq_cluster_engine_v5.py`, line 525 ```python confirmations['imb'] = imb_z > CFG.imb_z_threshold ``` **Problem:** `imb_z` is 0.00 on every entry (taker_buy_volume is zero on entry candles). With `imb_z_threshold=1.0`, this should always evaluate to `False`. But telemetry shows it passing 15/18 times (83%). **Root cause:** The `imb_z` computation at lines 512-520 has a fallback path: ```python if len(taker_buys) >= CFG.z_lookback and np.any(taker_buys[-CFG.z_lookback:] > 0): taker_ratios = taker_buys / np.maximum(volumes, 1e-10) imb_z = _z_score(taker_ratios, CFG.z_lookback) else: # Fallback: price-position relative to bar midpoint, normalized by ATR mid = (highs[-1] + lows[-1]) / 2 imb_z = (closes[-1] - mid) / (atr + 1e-10) ``` When `taker_buy_volume` is all zeros (which is the common case), the fallback computes a price-position metric that has nothing to do with taker imbalance. This fallback value is often > 1.0, causing the gate to pass. **Impact:** The imb gate is a free pass on 83% of entries. Combined with body (89%) and momentum (78%), trades pass 3/6 gates without any real confirmation. **Fix:** Either: 1. Remove the fallback â€" if taker data is zero, imb_z should be 0.0 (gate fails) 2. Or gate the entire imb confirmation on `has_taker` (skip the gate entirely when no taker data) ### ðŸŸ HIGH: Breakout Gate Never Fires **File:** `engines/liq_cluster_engine_v5.py`, line 524 ```python confirmations['breakout'] = closes[-1] > range_high ``` **Problem:** 0 of 18 post-loosening trades broke above the 60-bar range high. The system is buying into resistance, not breaking out. **Root cause:** In chopy markets, price rarely breaks above the 60-bar high. The range is too wide. **Impact:** The most important confirmation for a breakout strategy is never contributing. **Fix options:** 1. Shorten `range_lookback` from 60 to 30 2. Change to `closes[-1] > range_high * 0.995` (allow near-breakout) 3. Accept that breakout-as-defined isn't happening and the system is really momentum-based ### ðŸŸ¡ MEDIUM: FLAT_RISK_PCT NameError (V6.0 bug, may be fixed) **File:** `engines/liq_cluster_engine_v5.py` V6.0 introduced `FLAT_RISK_PCT` reference that wasn't imported, causing `NameError` on every signal. Was fixed to `BASE_RISK_PCT`. Verify this is

still correct. ### ðŸŸ¡ MEDIUM: bars_held Double Increment (FIXED in V6.0) Was fixed but worth verifying: the runner increments `candles_held` AND the engine increments `bars_held`. Only one should increment. Verify line ~581 in engine doesn't have `st.bars_held += 1`. ### ðŸŸ¢ LOW: Dead Code - `models.py` has 13-state `PositionState` machine â€” completely unused - `DECILE_EXITS` has entries for D4 and D10 â€” rejected but definitions remain - `btc_aligned` column in DB schema but never used --- ## 6. Live Performance Analysis ### Overall (149 closed trades) | Metric | Value | |--------|-------| | Total PnL | -28.25R | | Win Rate | 45.0% (67/149) | | Avg per trade | -0.190R | | Profit Factor | 0.58 | | Max drawdown | Not computed (equity snapshots exist) | ### By Exit Reason | Exit | Count | Total R | Avg R | |------|-------|---------|-------| | stop_loss | 48 | -46.15R | -0.96R | | struct_trail | 38 | +14.89R | +0.39R | | vol_trail | 21 | +6.35R | +0.30R | | time_stop | 26 | +1.40R | +0.05R | | early_cut | 15 | -6.17R | -0.41R | | cascade_deactivated | 1 | +1.42R | +1.42R | **Key insight:** Stop losses are -46.15R on 48 trades. That's the entire problem. Winners via trails are +22.64R on 59 trades. The exits work â€” the entries are wrong. ### By Decile | Decile | Trades | Total R | Avg R | WR | Status | |--------|--------|---------|-------|----|--------| | D1 | 72 | -15.67R | -0.218R | 47% | â�Œ Negative, but keep per Martin | | D2 | 18 | -4.55R | -0.253R | 44% | âš ï¸� Small sample | | D3 | 9 | -4.38R | -0.487R | 33% | ðŸš« Rejected | | D5 | 4 | +1.81R | +0.453R | 75% | âœ… Best edge | | D6 | 9 | -2.35R | -0.261R | 44% | âš ï¸� Small sample | | D7 | 13 | -3.34R | -0.257R | 31% | â�Œ Negative, but keep per Martin | | D8 | 12 | +2.91R | +0.242R | 58% | âœ… Solid | | D9 | 12 | -2.68R | -0.224R | 33% | â�Œ Negative, but keep per Martin | ### By Symbol (Top/Bottom 5) **Best:** | Symbol | Trades | Total R | WR | |--------|--------|---------|----| | DASHUSDT | 11 | +2.76R | 64% | | VIRTUALUSDT | 2 | +0.92R | 50% | | SOLUSDT | 12 | +0.77R | 58% | | FETUSDT | 12 | +0.63R | 67% | | XRPUSDT | 5 | +0.22R | 60% | **Worst:** | Symbol | Trades | Total R | WR | |--------|--------|---------|----| | ZECUSDT | 12 | -5.29R | 33% | | NEARUSDT | 22 | -4.41R | 50% | | LTCUSDT | 4 | -2.84R | 0% | | QNTUSDT | 8 | -2.51R | 38% | | UNIUSDT | 7 | -2.45R | 29% | ### Experimental vs Proven | Type | Trades | Total R | WR | |------|--------|---------|----| | Proven | 115 | -19.19R | 47% | | Experimental | 34 | -9.06R | 32% | --- ## 7. Confirmation Gate Analysis ### Post-Loosening Gate Pass Rates (18 trades) | Gate | Pass Rate | Notes | |------|-----------|-------| | body | 89% | 0.50 threshold too easy | | imb | 83% | **BUG** â€” imb_z=0.00 but gate passes | | momentum | 78% | Close > EMA20, easy in any uptrend | | impulse | 56% | 0.20% move, moderate | | vol | 17% | vol_z > 2.0, only 3/18 pass | | breakout | 0% | **Never fires** â€” no trade breaks 60-bar high | ### What the 18 Post-Loosening Trades Actually Passed Every trade passed on: **body + imb + momentum** (the 3 free gates). vol and breakout are the only real filters, and breakout never fires. ### Old vs New Thresholds | Gate | V6.3 (old) | V6.4 (new) | Problem | |------|-----------|-----------|---------| | vol_z | > 3.0 | > 2.0 | Still only 17% pass â€” too tight for chop | | imb_z | > 2.0 | > 1.0 | **Bug makes this irrelevant** â€” always passes | | body | â‰¥ 0.60 | â‰¥ 0.50 | 89% pass â€” too easy | | impulse | â‰¥ 0.30% | â‰¥ 0.20% | 56% pass â€” moderate | | min_confirms | 4/6 | 3/6 | Too easy with 3 free gates | --- ## 8. Stop Loss Analysis ### The -1R Problem - 48 stop losses, -46.15R total, avg -0.96R - 42 of 48 (93%) hit at almost exactly -1R - Avg 134 bars held before stop (11 hours) - The stop is set at 2.5 ATR and the trail never loosens enough ### R-Path Analysis (20 trades with telemetry) Of 20 stop-loss trades with full R-path data: - **11 (55%) never went positive** â€”

dead on arrival, nothing could save them - **5 (25%) flickered positive but < 0.3R** â€” brief hope, then died - **4 (20%) reached +0.3R+ then crashed back** â€” LTC (0.66Râ†’-1.05R), QNT (0.35Râ†’-1.05R), AAVE (0.31Râ†’-1.04R), INJ (1.50Râ†’+0.12R, this one survived) ### Winner Comparison Of 11 winners with R-path data: - **9 of 11 went negative at some point** before recovering - Deepest drawdown any winner survived: **-0.78R** (ZEC D2, closed +0.37R) - Avg minimum R among winners: **-0.46R** - Winners routinely gave back 0.5â€“1.7R from peak ### Simulated Stop Improvements | Option | Trades Helped | Trades Hurt | Net | |--------|-------------|------------|-----| | Widen stop 2.5â†’3.5 ATR | 4 saved | 16 lose -1.4R instead of -1.0R | **-2R** â�Œ | | Stop delay at +0.3R | 4 saved to breakeven | 16 unchanged | **+4R** âœ… | | Both combined | 4 saved | 16 unchanged | **+4R** âœ… | **Recommendation:** Stop delay at +0.3R (move stop to breakeven once trade reaches +0.3R) saves +4R with no downside. Wider stops actually hurt because most trades never show life. --- ## 9. Recommendations for Next AI ### Immediate Actions (Do First) 1. **REVERT V6.4 confirmation loosening** â€” go back to vol_z>3.0, imb_z>2.0, bodyâ‰¥0.60, impulseâ‰¥0.30, min_confirms=4. The loosening produced -11.16R on 18 trades. 2. **FIX the imb gate bug** â€” when taker_buy_volume is zero, imb_z should be 0.0 (gate fails), not the price-position fallback. This is the most impactful single fix. 3. **Investigate the breakout gate** â€” 0/18 trades break above 60-bar high. Consider shortening range_lookback to 30, or accept the system is momentum-based, not breakout-based. 4. **Implement stop delay** â€” once trade reaches +0.3R, move stop to breakeven. Saves +4R with no downside. ### Short-Term (This Week) 5. **Cut negative-experimental pairs:** LTC (-2.84R, 0% WR), HYPE (-2.14R, 0% WR), AVAX (-2.25R, 25% WR) 6. **Promote positive-experimental pairs:** VIRTUAL (+0.92R) â€” small sample but promising 7. **Session-based sizing:** London (08-16 UTC) is the only profitable session (59% WR, +1.21R). NY (16-24 UTC) is -10.64R. Consider reducing size 50% during NY. 8. **Symbol pruning:** ZEC (-5.29R), NEAR (-4.41R), QNT (-2.51R) are bleeding. Consider removing or reducing allocation. ### Medium-Term (Wait for Data) 9. **Keep D1, D7, D9 active** â€” Martin's explicit instruction. Wait for 200+ total trades before deactivating any decile. 10. **Wait for D5/D8 sample to grow** â€” these are the only deciles with positive expectancy. Need 20+ trades each to confirm. 11. **Pair expansion** â€” staged rollout: 15 pairs at a time at 0.5% experimental risk, gate on expectancy after 50+ trades each. ### What NOT to Do - Do NOT deactivate deciles (Martin's rule) - Do NOT touch production code without explicit approval - Do NOT push to GitHub without Martin's OK (learned the hard way) - Do NOT optimize for backtest PF â€” focus on live expectancy - Do NOT add random indicators â€” the 3-play framework is sound --- ## 10. Current Configuration Reference ### Engine Parameters (v6.4 â€” RECOMMEND REVERT) ```python # Cascade detection liq_lookback = 90 # days (reverted from 30) min_cascade_strength = 0.10 min_cascade_imb = 0.00 # disabled (was 0.30, removed in v6.3.2) # Entry confirmations (V6.4 â€” RECOMMEND REVERT to v6.3 values) range_lookback = 60 imb_z_threshold = 1.0 # v6.3: 2.0 vol_z_threshold = 2.0 # v6.3: 3.0 body_strength_min = 0.50 # v6.3: 0.60 impulse_min_pct = 0.20 # v6.3: 0.30 min_confirmations = 3 # v6.3: 4 z_lookback = 100 ema_period = 20 # Risk initial_stop_atr = 2.5 atr_period = 14 BASE_RISK_PCT = 0.04 # 4% (was FLAT_RISK_PCT, fixed in v6.0) # Selectivity cooldown_bars = 36 no_reentry_after_stop = True max_consecutive_stops = 3 stop_cooldown = 288 # 24h (was 999999, fixed in v6.0) # Rejected deciles TRADE_DECILES = {1, 2, 5, 6, 7, 8, 9} # D3, D4, D10 rejected # Early cut (v6.3 widened) early_cut_r_threshold = -0.5 # was -0.3 early_cut_mfe_threshold = 0.1 # was 0.3 ``` ### Decile Exit Parameters | Decile | vol_trail_atr | struct_lookback | max_hold_bars | decay_enabled | decay_start_bar | decay_min_r | |--------|-------------|----------------|--------------|--------------|----------------|-------------| | D1 | 3.0 | 48 | 500 | False | 999 | 999 | | D2 | 3.0 | 48 | 500 | False | 999 | 999 | | D5 | 2.0 | 12 | 288 | True | 15 | 1.5 | | D6 | 2.0 | 12 | 288 | True | 12 | 1.5 | | D7 | 2.5 | 36 | 358 | True | 20 | 2.0 | | D8 | 2.5 | 36 | 358 | True | 20 | 2.0 | | D9 | 1.5 | 8 | 100 | True | 8 | 1.5 | ### Symbol Tiers **Tier A (Proven, 35 symbols):** Full pipeline, Coinalyze-verified, backtested **Tier B (Proven additions):** Added in v6.2-v6.3 **Tier C (Experimental, 22 symbols):** WebSocket-only, 0.5% risk, gate on expectancy ### Key Files | File | Path | Purpose | |------|------|---------| | Engine | `engines/liq_cluster_engine_v5.py` | All trading logic | | Runner | `tools/v5_forward_test.py` | Live execution | | Config | `config/v5_forward_test.yaml` | Symbol/param config | | Telemetry | `v6_telemetry.py` | R-path, shadow exits | | Paper DB | `storage/v5_forward_test.db` | Trades, positions, state | | Telemetry DB | `storage/v6_telemetry.db` | R-path, entries, exits | ### Git Tags | Tag | Commit | Date | Description | |-----|--------|------|------------| | v5.2 | `ab5237c` | 2026-05-22 | Coinalyze â†’ Binance migration | | v5.3 | `ffb2f76` | 2026-05-23 | WebSocket liq feed | | v6.0 | `cb03ba7` | 2026-05-23 | Full audit fix (18 bugs) | | v6.2 | `dde5950` | 2026-05-24 | D3 rejection, directional filter, early cut | | v6.3 | `230c984` | 2026-05-24 | Tier 2 experimental pairs (50 total) | | v6.4 | `456dbed` | 2026-05-25 | Loosened confirmations (REVERT RECOMMENDED) | --- ## Appendix A: Session Performance | Session | UTC | Trades | WR | Total R | |---------|-----|--------|----|---------| | London | 08-16 | â€” | 59% | +1.21R | | Asian | 00-08 | â€” | 41% | -4.97R | | NY | 16-24 | â€” | 43% | -10.64R | 22:00 UTC is the worst hour: 11 entries, 27% WR, -7.10R. ## Appendix B: Kelly Analysis - Full Kelly: 8.8% (on V6.2 data without D3) - Half-Kelly: 4.4% - Current sizing: ~4% flat (half-Kelly) - D1 Kelly: 21.3% (but negative expectancy in choppy regime) - D5 Kelly: +48% (positive, but only 4 trades) - D8 Kelly: +17.4% (positive, 12 trades) ## Appendix C: Key Metrics to Monitor 1. **Expectancy per trade** â€” currently -0.19R (needs to be >0) 2. **Stop loss rate** â€” currently 32% (48/149), should be <25% 3. **D5/D8 trade count** â€” need 20+ each to confirm edge 4. **Breakout gate pass rate** â€” currently 0%, needs investigation 5. **imb gate correctness** â€” verify it's actually checking taker imbalance --- *End of report. For questions, check the git log at each tag. The commit messages are detailed and explain the reasoning behind each change.*